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Inventor(s)	WATANABE; Masayoshi et al.

ELECTRICITY-DRIVEN VEHICLE

Abstract

An electricity-driven vehicle includes: a floor panel; a deck board; a battery loading portion; one or more batteries of a replaceable type, that are inserted into and removed from the battery loading portion from a side of the vehicle; a loading port; and a lid that opens and closes the loading port. A marginal region is present between one end of the floor panel in a vehicle width direction, and the loading port. The deck board includes: a deck board body; and an overhang portion which extends to an outer side in the vehicle width direction in relation to the battery loading portion, so as to cover the marginal region from above the marginal region. The overhang portion is movable to a retracted position at which an area above the marginal region is opened.

Inventors: **WATANABE; Masayoshi (Nagoya-shi, JP), SAITO; Kan (Ikeda-shi, JP), TSUJITA; Toshiya (Ikeda-shi, JP), NAKAMURA; Naohiro (Ikeda-shi, JP), KAWAMOTO; Yasuo (Ikeda-shi, JP), OGAMI; Hideki (Ikeda-shi, JP)**

Applicant: **TOYOTA JIDOSHA KABUSHIKI KAISHA** (Aichi-ken, JP); **ISUZU MOTORS LIMITED** (Kanagawa, JP)

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION
[0001] This application claims priority to Japanese Patent Application No. 2024-029867 filed on Feb. 29, 2024, which is incorporated herein by reference in its entirety including the specification, claims, drawings, and abstract.
TECHNICAL FIELD
[0002] The present disclosure relates to an electricity-driven vehicle, equipped with a replaceable battery.
BACKGROUND
[0003] In the related art, an electricity-driven vehicle is widely known, in which a motor which is a motive power source, and a battery which supplies electric power to the motor are provided on a vehicle. The battery equipped in the electricity-driven vehicle is a secondary battery which can be charged and discharged, and is charged by an external power source as necessary.
[0004] A configuration is being proposed which employs a replaceable battery, which can be attached to and detached from the electricity-driven vehicle, in order to effectively utilize the electricity-driven vehicle even during charging of the battery. In a structure with a replaceable battery, when a state of charge of the battery equipped on the electricity-driven vehicle becomes low, the battery is removed from the electricity-driven vehicle, and is replaced with another battery which is charged in advance. The battery which is removed from the electricity-driven vehicle is then charged outside the electricity-driven vehicle. Because the electricity-driven vehicle can travel even during the charging of the removed battery, the electricity-driven vehicle can be more effectively utilized.
[0005] JP 2012-151916 A discloses a technique in which a replaceable battery is placed at a lower side of a floor panel of a vehicle. In the technique of JP 2012-151916 A, a loading port of the battery is positioned at an end of the electricity-driven vehicle in a vehicle width direction. In this case, when the vehicle collides at the side, impact of the collision is immediately input to the battery, and, thus, the battery may be damaged. Thus, a configuration may be considered in which a slight marginal region is provided between the loading port of the battery and the end of the vehicle in the vehicle width direction.
[0006] Normally, a panel member which is called a floor panel or a deck board is placed at an upper side of the battery. When the marginal region is covered by the panel member from above the marginal region, it becomes more difficult for the user to view a periphery of the loading port, resulting in reduction of workability of the battery replacement. On the other hand, if a portion of the panel member which overlaps the marginal region is cut out, because baggage cannot be loaded at the cut-out portion, a loading efficiency of the baggage is reduced.
[0007] An advantage of the present disclosure lies in provision of an electricity-driven vehicle which can achieve both superior loading efficiency of the baggage and superior workability of replacement of the battery.
SUMMARY
[0008] According to one aspect of the present disclosure, there is provided an electricity-driven

vehicle comprising: a floor panel; a deck board that is placed at an upper side of the floor panel; a battery loading portion that is placed between the floor panel and the deck board; one or more batteries of a replaceable type, that are inserted into and removed from the battery loading portion from a side of the vehicle; a loading port that is provided at one end of the battery loading portion in a vehicle width direction; and a lid that opens and closes the loading port, wherein a marginal region is present between one end of the floor panel in the vehicle width direction and the loading port, the deck board comprises: a deck board body; and an overhang portion which is placed at one end of the deck board in the vehicle width direction, and which extends to an outer side in the vehicle width direction in relation to the battery loading portion, so as to cover the marginal region from above the marginal region, and the overhang portion is movable to a retracted position in which an area above the marginal region is opened.
[0009] With the above-described structure, because baggage can be loaded also above the marginal region, the loading efficiency of the baggage is improved. In addition, with the overhang portion being moved to the retracted position, the battery can be easily replaced without the field of view being obstructed by the overhang portion.
[0010] In this case, the electricity-driven vehicle may further comprise a pair of side members that are placed with a spacing in the vehicle width direction, the battery loading portion may be placed within a width between the pair of side members, and the lid may open the loading port by swinging downward about an axis near a lower end of the loading port.
[0011] By employing a configuration in which the lid swings downward when the loading port is opened, obstruction of the field of view of the user by the lid when the lid is opened can be prevented. As a result, the user can easily replace the battery.
[0012] The overhang portion may move to the retracted position by sliding in a direction of a plane of the deck board.
[0013] According to the above-described structure, the overhang portion can be moved to the retracted position even in a state in which a part of the baggage is loaded on the overhang portion. As a result, the user can more easily replace the battery.
[0014] The overhang portion may be connected to the deck board body in a swingable manner, and may move to the retracted position by swinging upward.
[0015] With the above-described structure, the structure of the overhang portion can be simplified.
[0016] The deck board may further comprise a support member which is formed from a rigid structure, and which supports the overhang portion from below the overhang portion.
[0017] With the above-described structure, deformation of the overhang portion can be effectively prevented, and, thus, baggage can be stably disposed on the overhang portion.
[0018] According to an aspect of the present disclosure, superior loading efficiency of the baggage and superior workability of replacement of the battery can both be achieved.

Description

BRIEF DESCRIPTION OF DRAWINGS
[0019] Embodiment(s) of the present disclosure will be described based on the following figures, wherein:
[0020] FIG. **1** is a perspective diagram showing a region near a rear side door of an electricity-driven vehicle, viewed from outside the vehicle;
[0021] FIG. **2** is a schematic plan view of the electricity-driven vehicle;
[0022] FIG. **3** is a cross-sectional diagram along an A-A line of FIG. **2**;
[0023] FIG. **4** is a cross-sectional diagram along a B-B line of FIG. **2**;
[0024] FIG. **5** is a schematic diagram showing a relative placement of a deck board and a battery loading portion; and

[0025] FIG. **6** is a diagram showing another example of the deck board.

DESCRIPTION OF EMBODIMENTS

[0026] A structure of an electricity-driven vehicle **10** according to an embodiment of the present disclosure will now be described with reference to the drawings. FIG. **1** is a perspective diagram showing a region near a rear side door **12** of the electricity-driven vehicle **10**, viewed from outside the vehicle. FIG. **2** is a schematic plan view of the electricity-driven vehicle **10**. FIG. **3** is a cross-sectional diagram along an A-A line of FIG. **2**, and FIG. **4** is a cross-sectional diagram along a B-B line of FIG. **2**. FIG. **5** is a schematic diagram showing a relative placement of a deck board **30** and a battery loading portion **40**. In the drawings, “Fr”, “Up”, and “Rh” respectively refer to a front side, an upper side, and a right side of the electricity-driven vehicle **10**.

[0027] The electricity-driven vehicle **10** comprises a traveling motor (not shown), and a battery **44** which supplies electric power to the traveling motor. The electricity-driven vehicle **10** exemplified below is a commercial vehicle for transporting baggage. A rear seat of the electricity-driven vehicle **10** is removed, in order to improve a loading efficiency of the baggage. Therefore, an entirety of a rear side of a front seat **29** (not shown in FIG. **1**; refer to FIG. **3**) can be used as a baggage compartment. On a side of the electricity-driven vehicle **10**, a front side door **11** and the rear side door **12** are placed, aligned in a front-and-rear direction. As shown in FIG. **2**, the front side door **11** is a hinge door which swings about an axis extending in a vertical direction, and the rear side door **12** is a sliding door which slides in a vehicle front-and-rear direction. The rear side door **12** opens and closes a rear side door opening **14**.

[0028] The battery **44** is a replaceable battery which can be easily replaced in the electricity-driven vehicle **10**. The battery **44** is formed by housing a battery body formed from a plurality of unit cells in a housing. The housing of the battery **44** has, for example, a parallelepiped shape elongated in a vehicle width direction. A battery-side connector is provided at an end of the battery **44**. When the battery-side connector is connected to a vehicle-side connector **46** to be described below, an electric power system equipped on the vehicle and the battery **44** are electrically connected with each other. The battery body is a secondary battery (for example, a lithium ion battery or the like) which can be charged and discharged. Therefore, the battery **44** which is removed from the electricity-driven vehicle **10** can be charged externally to the vehicle.

[0029] By setting the battery **44** as a replaceable battery in this manner, the usage efficiency of the electricity-driven vehicle **10** can be improved, and a demand peak of commercial electric power can be reduced. That is, when the battery **44** is of the replaceable type, after the battery **44** is removed from the electricity-driven vehicle **10**, another battery **44** which is already charged can be mounted on the electricity-driven vehicle **10**. Further, with such a configuration, because the electricity-driven vehicle **10** can travel even during a period in which a part of the battery **44** is charged, the usage efficiency of the electricity-driven vehicle **10** is improved. In addition, when the battery **44** is of the replaceable type, time of performing the charging operation can be freely selected, regardless of a usage situation of the electricity-driven vehicle **10**. Because of this, the battery **44** can be charged during a time period of low demand of the commercial electric power (for example, at late night), and the demand peak of the commercial electric power can be reduced.

[0030] Such a battery **44** is mounted at a location where a user can easily replace the battery and where loading of the baggage is not obstructed. In the case of the electricity-driven vehicle **10** of the present disclosure, a battery loading portion **40** at which the battery **44** is loaded is provided at an adjacent position of the rear side door opening **14**. The battery loading portion **40** will now be described in detail.

[0031] First, prior to the description of the battery loading portion **40**, a basic vehicle body structure of the electricity-driven vehicle **10** will be described. The electricity-driven vehicle **10** has a vehicle body structure approximately identical to that of existing vehicles. By using the vehicle body structure of the existing vehicles as is, components and manufacturing processes can be made common with the existing vehicles, resulting in reduction of development and manufacturing costs of the electricity-driven vehicle **10**. As shown in FIGS. **2** and **4**, the electricity-driven vehicle **10** comprises a pair of side members **24** placed with a spacing in the vehicle width direction, and a plurality of cross members **26** connecting the pair of side members **24**. Each of the side members **24** and the cross members **26** is a frame of the electricity-driven vehicle **10**.

[0032] As shown in FIG. **4**, at an upper side of the side member **24**, a floor panel **16** is provided. The floor panel **16** is a steel plate panel forming a floor surface of a vehicle cabin. The floor panel **16** is welded to the frame (side member **24** or the like) of the electricity-driven vehicle **10**. As shown in FIG. **3**, at an immediately rear side of the front seat **29**, a depressed portion **18** is present. The depressed portion **18** is a portion which extends in the vehicle width direction and which is depressed in a groove shape. In the case of the existing vehicles, a rear seat is placed at an immediately rear side of the depressed portion **18**. As described above, in the case of the electricity-driven vehicle **10** of the present embodiment, the rear seat is removed.

[0033] At a laterally adjacent position of the depressed portion **18**, the rear side door opening **14** is placed. As shown in FIG. **4**, a lower end of the rear side door opening **14** is defined by a rocker **28**. The rocker **28** is one type of a frame extending in the vehicle front-and-rear direction. In the present embodiment, as shown in FIG. **4**, the battery loading portion **40** is provided at an upper side of the depressed portion **18**. In addition, as shown in FIG. **2**, the battery loading portion **40** is completely fitted within a width of the rear side door opening **14** in the front-and-rear direction. By employing such a configuration, the user can easily access the battery loading portion **40** through the rear side door opening **14**.

[0034] The battery loading portion **40** is a part at which the battery **44** is loaded. The battery loading portion **40** has, for example, a battery casing **42**. As shown in FIG. **4**, the battery casing **42** is a container which houses one or more batteries **44** (in the illustrated example configuration, three batteries **44**), and is a box approximately having a parallelepiped shape with one end in the vehicle width direction opened. The opening of the battery casing **44** at one end in the vehicle width direction serves as a loading port **48** through which the battery **44** is inserted and removed. When the battery **44** is to be replaced, the user accesses the battery casing **42** from a side of the electricity-driven vehicle **10** (in the illustrated example configuration, the left side), and inserts or removes the battery **44** via the loading port **48**. As described above, in the present embodiment, the rear side door **12** is the sliding door. By employing the sliding door, interference between the battery **44** to be inserted or removed and the rear side door **12** can be effectively prevented, and workability when the battery **44** is replaced can be improved.

[0035] The loading port **48** is opened and closed by a casing lid **50**. For example, the casing lid **50** swings with the center near a lower end of the loading port **48**. As shown in FIG. **4**, the casing lid **50** closes the loading port **48** when in a standing orientation approximately parallel with a perpendicular surface. When the casing lid **50** in the standing orientation swings downward, the loading port **48** is opened. By swinging the casing lid **50** about the lower end of the loading port **48** in this manner, a configuration is achieved in which, when the casing lid **50** is opened, the casing lid **50** does not interpose between eyes of the user (which are normally positioned above the loading port **48**) and the loading port **48**. With this configuration, the user can more easily see the periphery of the loading port **48**, and, as a consequence, the workability when the battery **44** is replaced can be improved. In addition, in the case of the illustrated example configuration, a dimension of the casing lid **50** in the up-and-down direction is significantly smaller than the dimension of the casing lid **50** in the front-and-rear direction. Thus, by employing a configuration where the casing lid **50** opens downward, as compared with a case where the casing lid **50** opens laterally, an amount of protrusion when the casing lid **50** is opened can be set small. However, the structure of the casing lid **50** is not limited to the illustrated example configuration, and may be suitably changed. For example, the casing lid **50** may be changed to a laterally opening type, a double door type, or a sliding type.

[0036] In the battery casing **42**, the connector **46** to be electrically connected to the battery **44** loaded in the battery casing **42** is provided. The battery **44** is electrically connected to an electric power system which is equipped in the vehicle, via the connector **46**.

[0037] As shown in FIGS. **2** and **4**, the battery loading portion **40** (that is, the battery casing **42**) is fitted within a width between two side members **24** in the plan view. With such a configuration, when the electricity-driven vehicle **10** collides at the side, the side member **24** receives the impact before the battery loading portion **40**. Because of this, damage to the battery **44** housed in the battery loading portion **40** is effectively prevented.

[0038] When the battery loading portion **40** is placed within the width between two side members **24**, a vacant space in which nothing is placed is formed between one end of the floor panel **16** in the vehicle width direction and the loading port **48**. In the following description, this vacant space between the floor panel **16** and the loading port **48** will be described as a "marginal space **17**".

[0039] As shown in FIG. **3**, a height of the battery casing **42** is shorter than a depth of the depressed portion **18**. The electricity-driven vehicle **10** further comprises a deck board **30** placed at an upper side of the floor panel **16** so as to cover and hide the depressed portion **18**. In other words, the battery loading portion **40** is placed in a space surrounded by the deck board **30** and the depressed portion **18**. As shown in FIGS. **2** and **3**, the deck board **30** forms a flat floor surface, extending from a rear side of the front seat **29** to a rear end of the vehicle. By providing such a deck board **30**, the entirety of the space behind the front seat **29** can be used as a baggage compartment, and thus, a transport efficiency of the baggage by the electricity-driven vehicle **10** can be improved.

[0040] The deck board **30** is not welded to any of the members of the electricity-driven vehicle **10**, and can be removed from the electricity-driven vehicle **10** without destroying a component of the electricity-driven vehicle **10**. By removing the deck board **30** from the electricity-driven vehicle **10** as necessary, maintenance or the like of the battery loading portion **40** can be easily performed.

[0041] As shown in FIGS. **2**, **4**, and **5**, the deck board **30** comprises a body **32** and an overhang portion **34**. The body **32** is a panel member which covers most part of the baggage compartment, except for the marginal region **17** described above. As shown in FIGS. **2** and **5**, on the body **32**, a cutout **33** is formed in an area which overlaps the marginal region **17** in a plan view.

[0042] The overhang portion **34** is a panel member which is placed at one end of the deck board **30** in the vehicle width direction, and which covers the marginal region **17** from above the marginal region **17**. In other words, the overhang portion **34** is a panel member placed at the cutout **33**. Therefore, as shown in FIG. **4**, the overhang portion **34** extends to an outer side in the vehicle width direction in relation to the battery loading portion **40**. In the example configuration of FIG. **4**, the overhang portion **34** is connected to the body **32** via a hinge mechanism **36**. With the hinge mechanism **36**, the overhang portion **34** is swingable about an axis extending in the vehicle front-and-rear direction. Further, the overhang portion **34** is displaceable between a normal position at which the marginal region **17** is covered from above the marginal region **17**, and a retracted position at which an area above the marginal region **17** is opened, through the swinging. In FIG. **4**, the overhang portion **34** at the retracted position is illustrated with a solid line, and the overhang portion **34** at the normal position is illustrated with two-dots-and-chain line.

[0043] As shown in FIGS. **4** and **5**, at an end of the cutout **33** at an outer side in the vehicle width direction, a support member **38** is provided, which is a rigid structure supporting the overhang portion **34** at the normal position from below the overhang portion **34**. The support member **38** is, for example, a metal bar-shaped member extending across the cutout **33** in the vehicle front-and-rear direction.

[0044] As already described, the overhang portion **34** is displaceable between the normal position at which the marginal region **17** is covered from above the marginal region **17**, and the retracted position at which the area above the marginal region **17** is opened. When the overhang portion **34** is positioned at the normal position, the baggage can be loaded also at the region above the marginal region **17**. With this configuration, the loading efficiency of baggage of the electricity-driven vehicle **10** can be improved. Further, in the normal position, the overhang portion **34** is supported by the support member **38**. Because of this, even when the baggage is heavy, flexure and damages of the overhang portion **34** can be prevented.

[0045] When the user replaces the battery **44**, the user moves the overhang portion **34** to the retracted position. With such a configuration, the user can well check the periphery of the loading portion **48** without being obstructed by the overhang portion **34**. Thus, with this configuration, the workability of the replacement of the battery **44** can be further improved.

[0046] There may be cases in which, during the replacement operation of the battery **44**, the overhang portion **34** experiences some impact or vibration, and falls from the retracted position to the normal position. In order to prevent such unintended falling of the overhang portion **34**, a latch mechanism which temporarily maintains a swinging angle of the overhang portion **34** may be provided on the hinge mechanism **36**. For example, the hinge mechanism **36** may be a ratchet hinge that maintains the swinging angle with a predetermined angle interval, from the normal position to the retracted position.

[0047] In the structure in which the overhang portion **34** swings by the hinge mechanism **36** as shown in FIG. **4**, it is necessary to completely move the baggage to the outer side of the overhang portion **34** before the overhang portion **34** is moved to the retracted position. In other words, in the structure of FIG. **4**, the user must move the baggage prior to the replacement of the battery **44**, which is cumbersome.

[0048] Thus, a configuration may be employed in which the overhang portion **34** can move to the retracted position without interfering with the baggage mounted on the deck board **30**. For example, as shown in FIG. **6**, a lower portion of the body **32** may be set as the retracted position. In this case, the overhang portion **34** moves to the retracted position through a sliding movement in the direction of the plane of the body **32**. In this case, for example, a slide rail (not shown) is fixed to a back surface of the body **32**, and a movable element (not shown) which moves along the slide rail is fixed to a front surface of the overhang portion **34**. In this manner, by slide-moving the overhang portion **34**, even when a part of the baggage extends into the marginal region **17**, the overhang portion **34** can be moved to the retracted position. With this configuration, the battery **44** can be well replaced without moving the baggage.

[0049] In the above description, the overhang portion **34** is mechanically connected to the body **32** via the hinge mechanism **36** or the slide rail, and is not completely separated from the body **32**. Alternatively, a configuration may be employed in which the overhang portion **34** is completely separable from the body **32**. For example, the overhang portion **34** may be a panel member mounted on a periphery of the body **32** and the support member **38** in a manner to bridge therebetween.

[0050] The structures described above are merely exemplary, and, so long as the electricity-driven vehicle has the structure described in Claim **1**, the other structures may be suitably changed. For example, in the above description, the rear side door **12** is described as a sliding door, but alternatively, the rear side door **12** may be a hinge door, similar to the front side door **11**. Further, the structure of the battery loading portion **40** may also be suitably changed. For example, the shape and the size of the battery loading portion **40**, and presence or absence of the casing lid **50** may be suitably changed.

REFERENCE SIGNS LIST

[0051] **10** electricity-driven vehicle, **11** front side door, **12** rear side door, **14** rear side door opening, **16** floor panel, **17** marginal region, **18** depressed portion, **24** side member, **26** cross member, **28** rocker, **29** front seat, **30** deck board, **32** body, **33** cutout, **34** overhang portion, **36** hinge mechanism, **38** support member, **40** battery loading portion, **42** battery casing, **44** battery, **46** connector, **48** loading port, **50** casing lid

Claims

1. An electricity-driven vehicle comprising: a floor panel; a deck board that is placed at an upper side of the floor panel; a battery loading portion that is placed between the floor panel and the deck board; one or more batteries of a replaceable type, that are inserted into and removed from the battery loading portion from a side of the vehicle; a loading port that is provided at one end of the battery loading portion in a vehicle width direction; and a lid that opens and closes the loading port, wherein a marginal region is present between one end of the floor panel in the vehicle width direction and the loading port, the deck board comprises: a deck board body; and an overhang portion which is placed at one end of the deck board in the vehicle width direction, and which extends to an outer side in the vehicle width direction in relation to the battery loading portion, so as to cover the marginal region from above the marginal region, and the overhang portion is movable to a retracted position at which an area above the marginal region is opened.
2. The electricity-driven vehicle according to claim 1, further comprising: a pair of side members that are placed with a spacing in the vehicle width direction, wherein the battery loading portion is placed within a width between the pair of side members, and the lid opens the loading port by swinging downward about an axis near a lower end of the loading port.
3. The electricity-driven vehicle according to claim 1, wherein the overhang portion moves to the retracted position by sliding in a direction of a plane of the deck board.
4. The electricity-driven vehicle according to claim 1, wherein the overhang portion is connected to the deck board body in a swingable manner, and moves to the retracted position by swinging upward.
5. The electricity-driven vehicle according to claim 1, wherein the deck board further comprises a support member which is formed from a rigid structure, and which supports the overhang portion from below the overhang portion.